Tutorial on AMP-PPI

This tutorial provides a step-by-step guide on predicting protein-protein interactions (PPI) between antimicrobial peptide (AMP) and their target proteins using AMP-PPI.

Upon visiting the AMP-PPI homepage, users can initiate the prediction process by clicking the "Prediction" button on the left side, as shown in Figure 1.

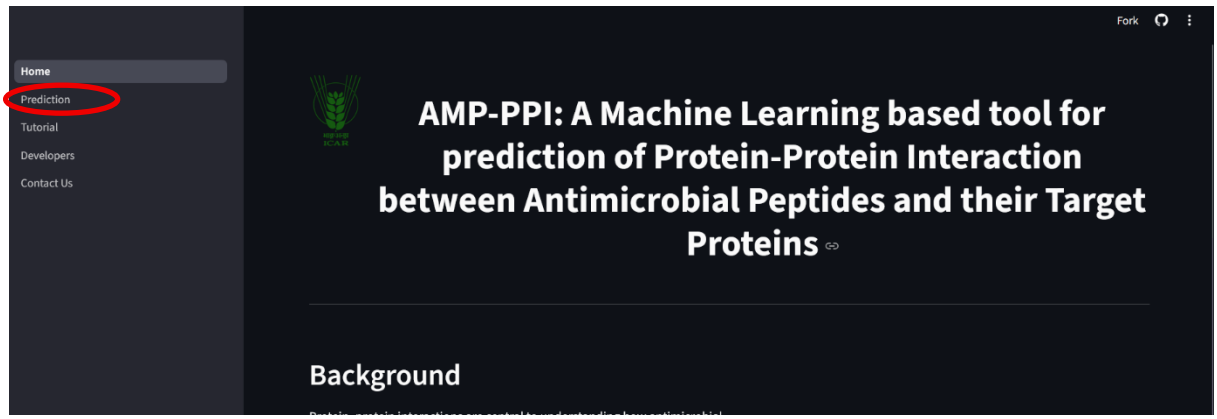


Figure 1: Homepage of AMP-PPI

Providing Input Sequences

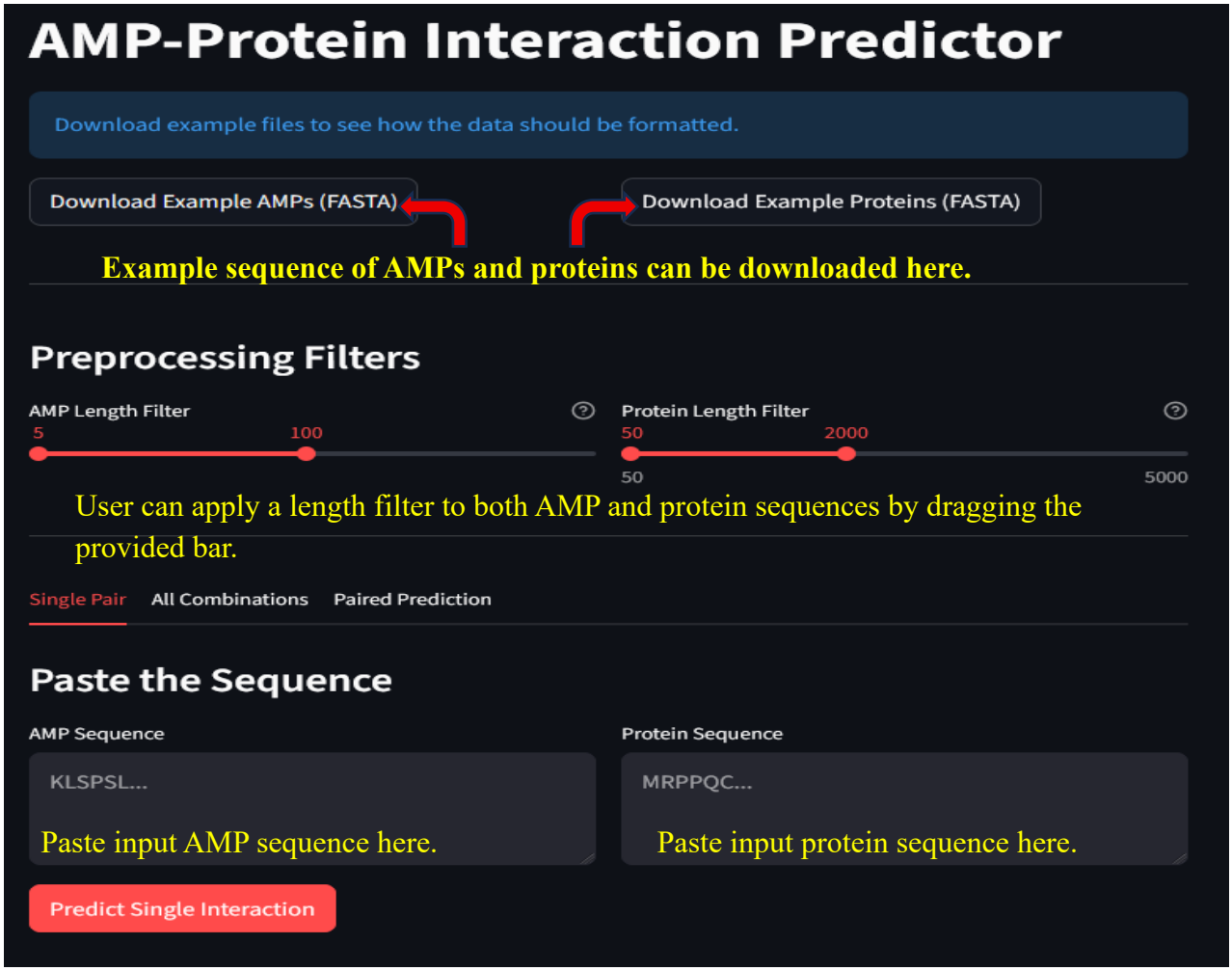
Users can input AMP and protein sequences in two ways:

- Paste the sequences directly into the respective text boxes.
- Upload sequences in FASTA format.

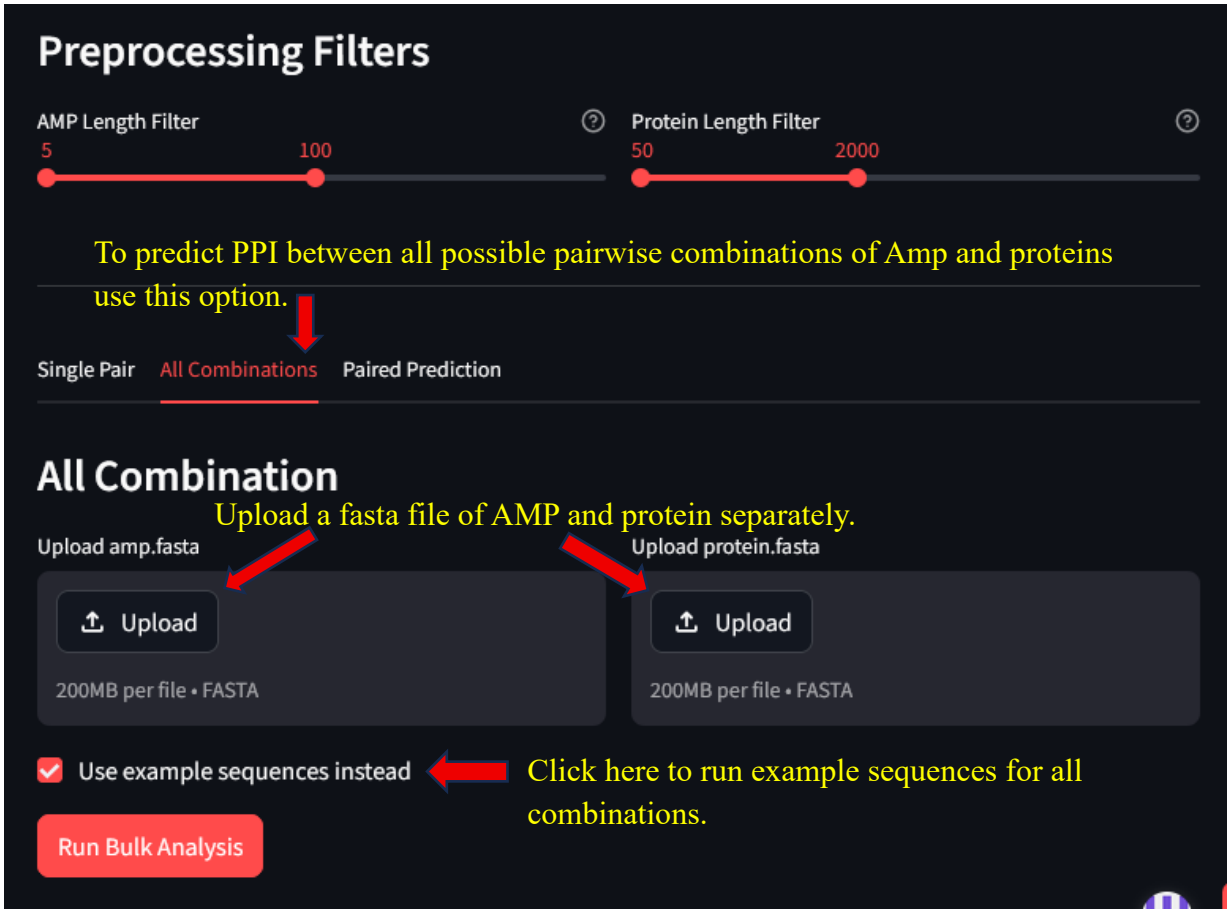
Additionally, users can:

- Provide pre-defined protein pairs for interaction prediction.
- Input a variable number of AMP and target proteins and generate all possible pairwise combinations for analysis (Figure 2)

a)


AMP-Protein Interaction Predictor
Download example files to see how the data should be formatted.
Download Example AMPs (FASTA)
Download Example Proteins (FASTA)
Example sequence of AMPs and proteins can be downloaded here.
Preprocessing Filters
AMP Length Filter
5
100
Protein Length Filter
50
2000
50
5000
User can apply a length filter to both AMP and protein sequences by dragging the provided bar.
Single Pair
All Combinations
Paired Prediction
Paste the Sequence
AMP Sequence
KLSPSL...
Paste input AMP sequence here.
Protein Sequence
MRPPQC...
Paste input protein sequence here.
Predict Single Interaction

b)


Preprocessing Filters
AMP Length Filter
5
100
Protein Length Filter
50
2000
To predict PPI between all possible pairwise combinations of Amp and proteins use this option.
Single Pair
All Combinations
Paired Prediction
All Combination
Upload a fasta file of AMP and protein separately.
Upload amp.fasta
Upload
200MB per file • FASTA
Upload protein.fasta
Upload
200MB per file • FASTA
Use example sequences instead
Click here to run example sequences for all combinations.
Run Bulk Analysis

c)

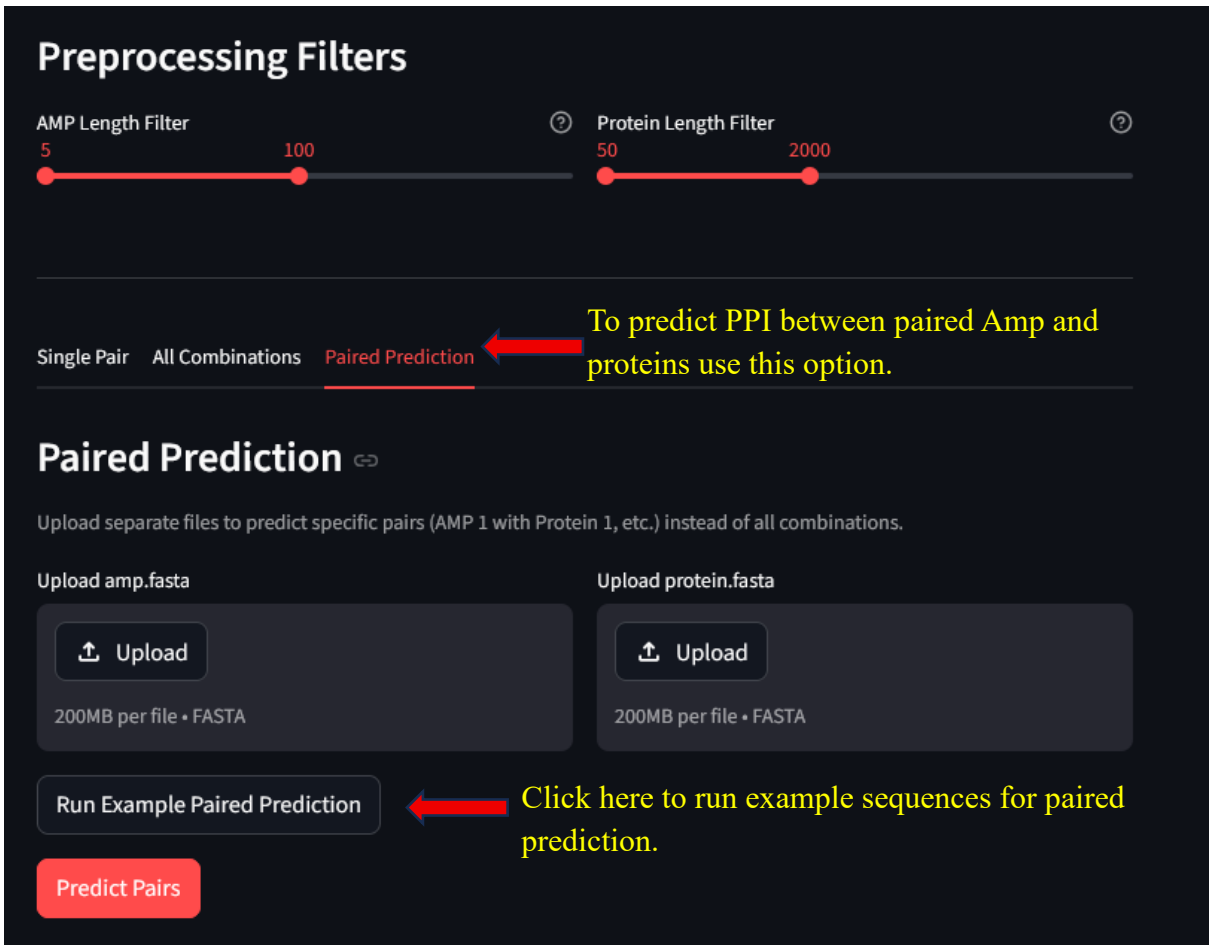


Figure 2: View of the prediction tab. a) Option to paste the sequence(s) in respective tab. b) and c) Option to upload the sequence(s) in fasta format.

Viewing and Interpreting Results

Predictions are displayed in a tabular format and can be downloaded for further analysis (Figure 3).

Prediction outcomes:

- ✓ 1 → Positive interaction predicted.
- ✓ 0 → No interaction predicted.

Each prediction includes an interaction probability score.

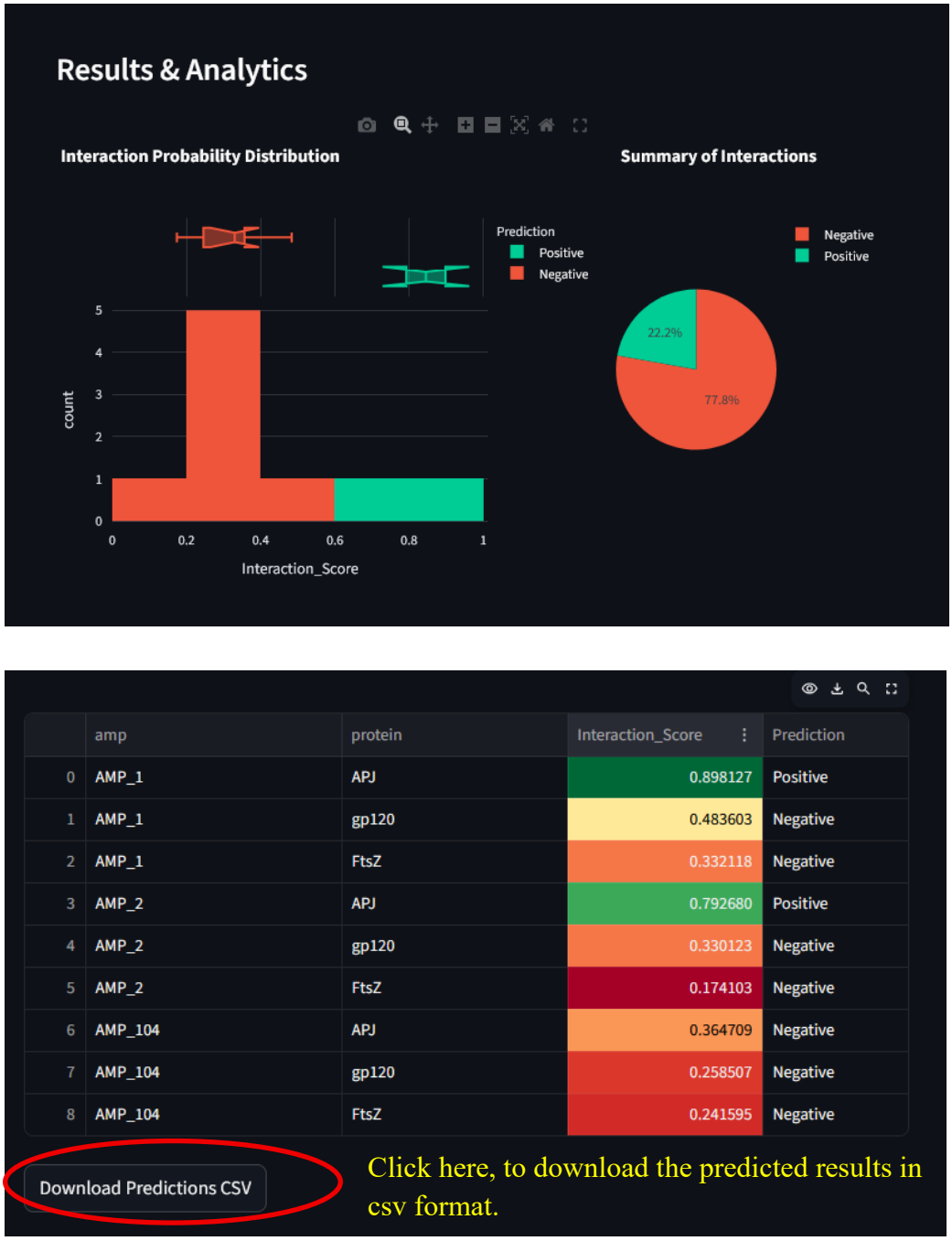


Figure 3: Result view.

In the given example, the result shows that the probability of interaction between the first and third given protein pairs is 100%. '1' in the 'Prediction' column means that the model predicted positive interaction. Whereas, second pair as well as the bottom five pairs have a higher probability of not interacting and hence the final prediction was in favor of negative interaction or no interaction represented by '0'.

Downloading and Analyzing Results

Users can download results for offline analysis. The downloaded results can be viewed in Excel or any spreadsheet tool (Figure 4).

	A	B	C	D	E
1	amp	protein	Interaction_Scor	Prediction	
2	AMP_1	APJ	0.898127458	Positive	
3	AMP_1	gp120	0.483603114	Negative	
4	AMP_1	FtsZ	0.332118339	Negative	
5	AMP_2	APJ	0.792680352	Positive	
6	AMP_2	gp120	0.330123147	Negative	
7	AMP_2	FtsZ	0.174102514	Negative	
8	AMP_104	APJ	0.364708776	Negative	
9	AMP_104	gp120	0.258507457	Negative	
10	AMP_104	FtsZ	0.241595325	Negative	
11					

Figure 4: Snapshot of the result